Matlab Stress Concentration Analysis in Plate with Two Elliptical Holes

For a simple circular hole in a uniaxially loaded plate, this stress concentration factor can be as high as three, meaning the peak stress at the edge of the hole is three times the farfield stress. These high-stress regions are critical because they are the most common sites for fatigue crack initiation, ultimately governing the component's service life.

Two primary methods have been established to mitigate this issue:

1. Shape Optimization: Modifying the shape of the main hole itself to reduce sharp stress gradients. The ideal shape is one where the stress is uniformly distributed along the boundary. [1]

2. Auxiliary Holes: Introducing smaller, strategically placed "stress-relieving" holes near the primary hole. These auxiliary holes act to intercept and redirect the flow of stress trajectories, smoothing them around the main discontinuity and thereby reducing the peak stress. [2]

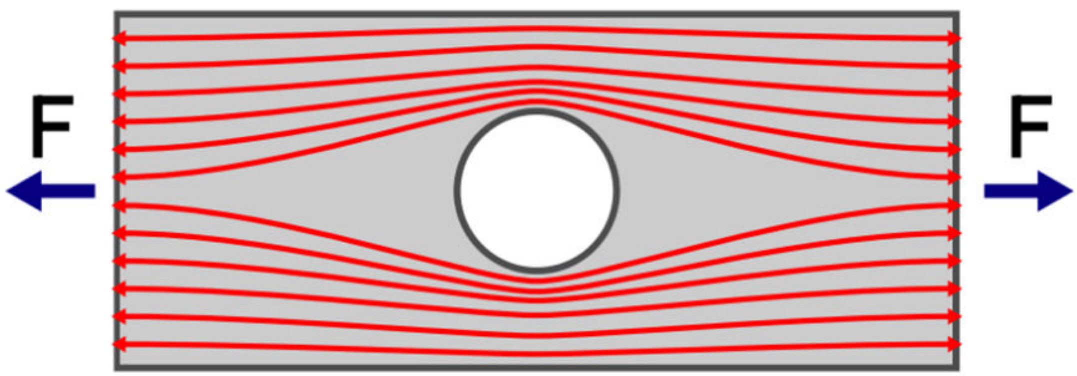

Fig: Plate with a Stress-Relieving Hole

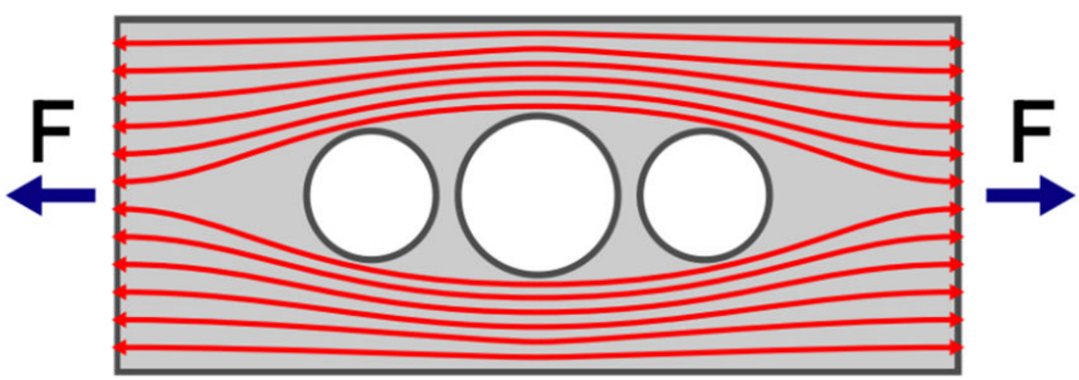

Fig: Plate with Auxiliary Stress-Relieving Holes

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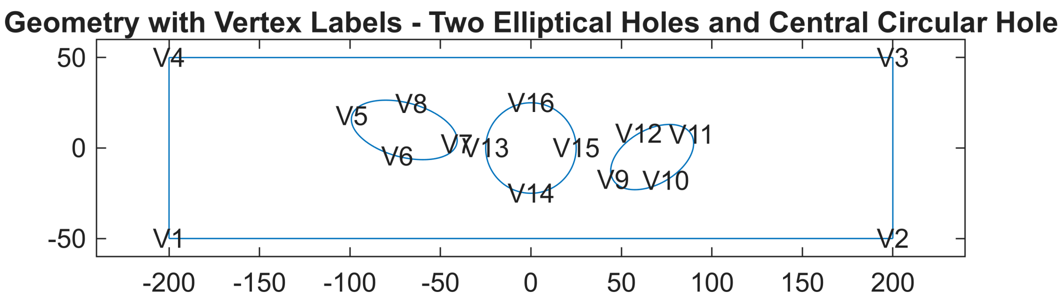
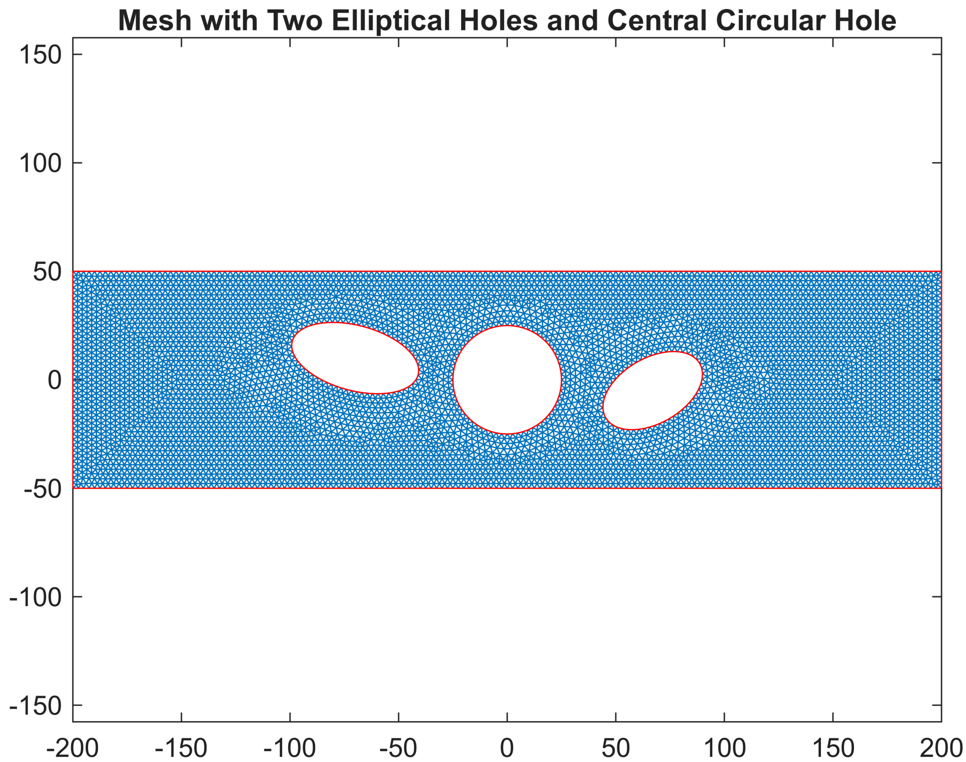
The following work will seek to analyze the stress concentration with fixed elliptical hole geometry..

Later on, it will expand to simultaneously optimize the shape of the main hole and the shape, size, and location of elliptical auxiliary holes to achieve the maximum possible reduction in stress concentration.

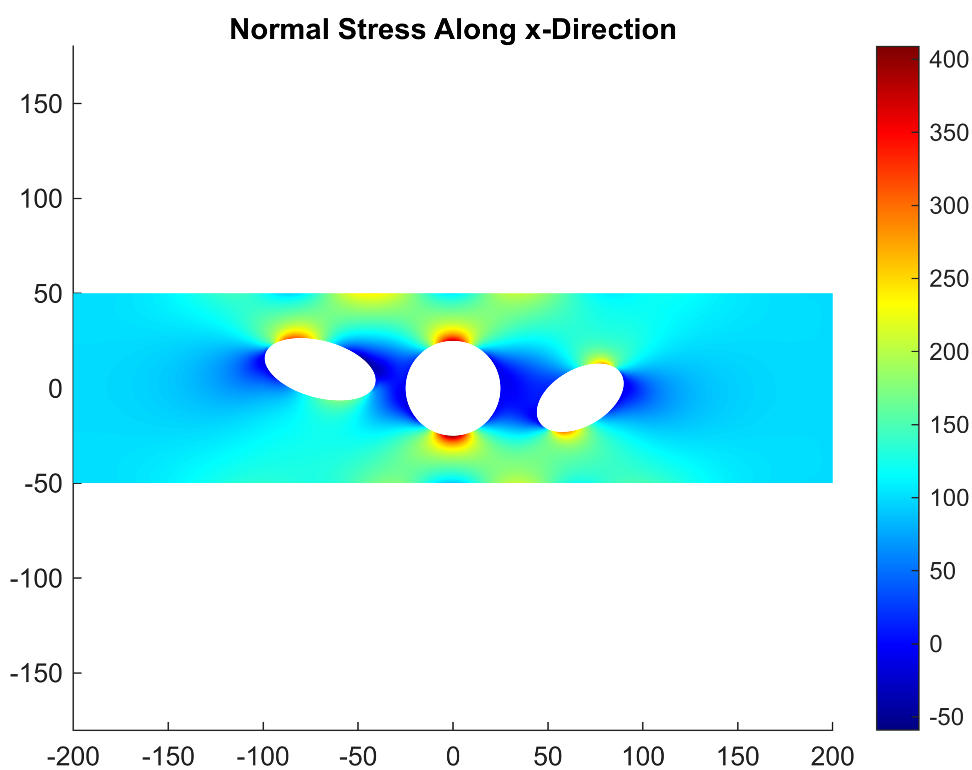
2-D plane-stress elasticity analysis with two elliptical holes

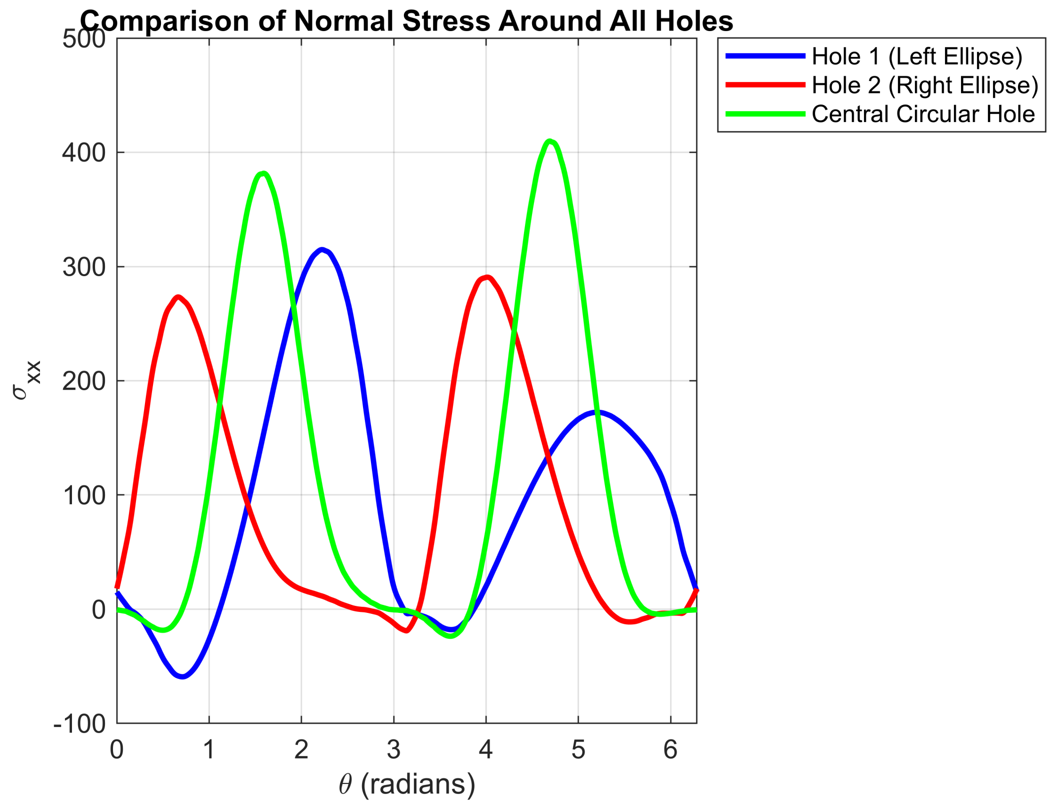


Mesh Generation



Stress Distribution Analysis



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Applied stress: 100
Ellipse 1 - Semi-axes: a=30.00, b=15.00, rotation=-0.26 rad
Maximum stress around Ellipse 1: 314.78
Stress concentration factor (Ellipse 1): 3.15
Ellipse 2 - Semi-axes: a=25.00, b=15.00, rotation=0.52 rad
Maximum stress around Ellipse 2: 290.67
Stress concentration factor (Ellipse 2): 2.91
Central Circular Hole - Radius: 25.00
Maximum stress around Central Hole: 409.81
Stress concentration factor (Central Hole): 4.10
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